EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Separation & Purification of organic compounds

1. For the Separation of two immiscible liquids which method (or apparatus) is used?
(A) Chromatography (B) Fractionating column
(C) Fractional distillation (D) Separating funnel / Decantation
CPO073

2. Aniline is usually purified by
(A) Chromatographic technique (B) Steam distillation
(C) By addition of oxalic acid (D) Fractional crystallization
CPO074

3. The most suitable method for separation of 1 : 1 mixture of ortho- and para- nitrophenols is
(A) Sublimation (B) Chromatography
(C) Crystallization (D) Steam distillation
CPO075

4. Distillation under reduced pressure is employed for:
(A) Benzene (B) Petrol
(C) Glycerol (D) Organic compound used in medicine
CPO076

Rankers Academy JEE

5. In sodium fusion test of organic compound the nitrogen of organic compound is converted into
(A) Sodamide (B) Sodium cyanide (C) Sodium nitrite (D) Sodium nitrate
CPO077

6. The latest technique used for purification for organic compounds is
(A) Chromatography (B) Sublimation
(C) Fractional distillation (D) Crystallisation
CPO078

7. In paper chromatography
(A) Mobile phase is liquid and stationary phase is solid
(B) Mobile phase is liquid and stationary phase is liquid
(C) Mobile phase is solid and stationary phase is solid
(D) Mobile phase is solid and stationary phase is liquid
CPO079

8. The Lassaigne's extract is boiled with dil. HNO_3 before testing for halogens because
(A) Silver halides are soluble in HNO_3 (B) Na_2S and NaCN are decomposed by HNO_3
(C) Ag_2S is soluble in HNO_3 (D) AgCN is soluble in HNO_3
CPO080

9. In a Lessaigne's test for nitrogen, the blue colour is due to the formation of
(A) Potassium ferricyanide (B) Sodium cyanide
(C) Sodium ferrocyanide (D) Ferri-ferro cyanide
CPO081

For More Join: @IITJEE_Advanced

10. Which of the following compounds does not show Lassaigne test for nitrogen?
(A) Urea (B) Hydrazine
(C) Phenyl hydrazine (D) Azo benzene

CPO082

11. When piece of human hair are heated strongly with soda lime, smell of ammonia can be detected. which one of the following conclusion can be drawn from this observation?
(A) Ammonia is present in human hair (B) Ammonium salt is present in hair
(C) Hair contains amino acids (D) None of the above

CPO083

12. The Beilstein test for organic compounds is used to detect
(A) Nitrogen (B) Sulphur (C) Carbon (D) Halogen

CPO084

13. Which of the following is the best scientific method to test presence of water in liquid?
(A) Smell (B) Taste
(C) Use of litmus paper (D) Use of anhydrous copper sulphate

CPO085

Quantitative Analysis

14. In Kjeldahl's method, nitrogen present is estimated as
(A) N_2 (B) NO (C) NH_3 (D) NO_2

CPO086

15. Liebig's method is used for the estimation of
(A) Nitrogen (B) Sulphur
(C) Carbon and hydrogen (D) Halogens

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CPO087

16. 0.765 gm of an acid gives 0.535 gm of CO_2 and 0.138 gm of H_2O. Then the ratio of percentage of carbon and hydrogen is
(A) 19 : 2 (B) 18 : 11 (C) 20 : 17 (D) 1 : 7

CPO088

17. Nitrogen is an organic compound can be estimated by
(A) Kjeldahl's method only (B) Duma's method only
(C) Both the methods (D) None of these methods

CPO089

18. Dumas method involves the determination of nitrogen content in the organic compound in form of
(A) NH_3 (B) N_2 (C) NaCN (D) $(NH_4)_2SO_4$

CPO090

Test of Functional Group

19. Phenol can be distiguished from ethanol by
(A) Tollen's reagent (B) Benedict's reagent
(C) $FeCl_3$ (D) Schiff 's base

CPO096

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Test of Functional Group

1. Which of the following will respond to Iodoform test?

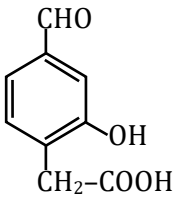
(A) $CH_3-\overset{\overset{O}{\|}}{C}-COOH$ (B) $CH_3-\overset{\overset{OH}{|}}{CH}-COOH$ (C) $CH_3-\overset{\overset{OH}{|}}{CH}-CH_3$ (D) CH_3CH_2OH

CPO098

2. Which of the following will not show iodoform test?

(A) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$ (B) $CH_3-\overset{\overset{O}{\|}}{C}-Cl$ (C) $CH_3-\overset{\overset{O}{\|}}{C}-NH_3$ (D) CH_3-COOH

CPO099

3. An organic compound has the structure  . It will give

(A) cerric ammonium nitrate test
(B) brick effervescence with sodium bicarbonate
(C) a characteristic colouration with neutral ferric chloride after decarboxylation and reduction by Clemenson's method
(D) Fehling's test

Bankers Academy JEE

CPO100

4. Compound A reacts with $CHCl_3$ and KOH and gives an offensive smelling compound. A can be

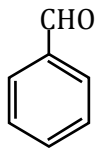
(A) Primary aliphatic amine (B) Primary aromatic amine
(C) Secondary amine (D)Tertiary amine

CPO101

5. HCOOH and HCHO may be distinguished by

(A) Tollen's test (B) Sodium bicarbonate test
(C) 2,4-DNP test (D)Benedict's test

CPO102

6.  and CH_3CHO can be distinguished By

(A) Tollen's test (B) Benedict'stest (C) Iodoform test (D) 2,4-DNP test

CPO103

7. Acetic acid and CH_3COCl can be distinguished by

(A) $NaHCO_3$ test (B) Na metal test
(C) Ester formation test (D)Br_2(aq.) test

CPO104

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

Quantitative Analysis

Paragraph for Question No. 8 to 12

The 0.2 g of anhydrous organic acid gave on combustion 0.04 g of water and 0.195 g of CO_2, The acid is a dibasic acid and 0.5 g of its silver salt leaves on ignition 0.355 g of silver.

8. The percentage of carbon in the compound is

(A) 50 (B) 52 (C) 26.6 (D) 90

CPO091

9. The percentage of hydrogen in the compound is

(A) 5.6 (B) 2.22 (C) 4.44 (D) 10

CPO092

10. The empirical weight of the compound is

(A) 90 (B) 100 (C) 10 (D) 45

CPO093

11. The molecular weight of the compound is

(A) 90 (B) 100 (C) 10 (D) 45

CPO094

12. The molecular formula of the compound is

(A) CHO (B) CHO_2 (C) $C_2H_2O_4$ (D) $C_3H_6O_2$

Rankers Academy JEE

CPO095

MATRIX MATCH TYPE QUESTIONS

Test of Functional Group

13. **Match the column-I and Column-II**

	Column I (reagents)		Column II
(a)	Tollen's reagent gives white precipitate with	(p)	Me–CH=CH–Me
(b)	$Br_2 + H_2O$ test given by	(q)	Me–C≡C–H
(c)	Product of reaction of acetylene with 1% $HgSO_4$ an Dil. H_2SO_4	(r)	$\overset{\ominus}{Cl}\ \overset{\oplus}{N}H_3 - NH_2$
(d)	Pd/H_2 reacts with	(s)	Me–C(=O)–H
		(t)	Me–C(=O)–Me

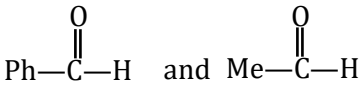
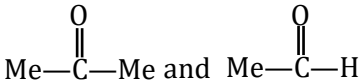
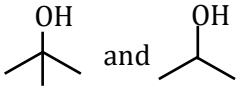
(A) (a→qr), (b→pq) ,(c→s), (d→pqst) (B) (a→qr), (b→q) ,(c→t), (d→pst)

(C) (a→pr), (b→pq) ,(c→p), (d→st) (D) (a→ts), (b→rs) ,(c→s), (d→qst)

CPO105

For More Join: @IITJEE_Advanced

14. Match the column-I and Column-II

	Column–I (Pair of compounds)		**Column–II (Test of identification)**
(a)		(p)	Tollen's reagent
(b)		(q)	Fehling's reagent
(c)		(r)	Iodoform test
(d)	Me–C≡C–H and Me–C≡C–Me	(s)	Victor Meyer's test
		(t)	Lucas test

(A) (a→q), (b→pq) ,(c→rt), (d→s)
(B) (a→qr), (b→pq) ,(c→rst), (d→p)
(C) (a→r), (b→st) ,(c→rs), (d→p)
(D) (a→pq), (b→rs) ,(c→st), (d→r)

CPO106

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For More Join: @IITJEE_Advanced

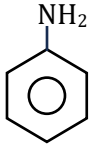
EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

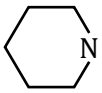
- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Qualitative Analysis

1. Amongst the following, the total number of compound which does not give Lassaigne's test for nitrogen.

I.  II. H_2N-CH_2-COOH III. NH_2-NH_2 IV. $Ph-N=N-Ph$ V. NH_2OH

VI. $NH_2-\overset{\overset{O}{\|}}{C}-NH_2$ VII. $Ph-NH-\overset{\overset{O}{\|}}{C}-Ph$ VIII. $PhN_2^{\oplus}Cl^{\ominus}$ IX. CH_3CHO X. CH_3CN

XI. 

CPO107

2. Amongst the following, the total number of compounds which are expected to give blood red colouration when $FeCl_3$ is added to their respective Lassaigne's extract is:

(1) thiourea (2) benzenesulphonic acid (3) sulphanilic acid

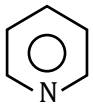
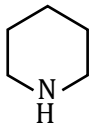
(4) diphenyl sulphide (5) urea (6) benzanilide

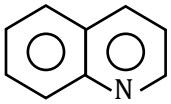
(7) semicarbazide hydrochloride (8) *p*-toluidine (9) *m*-dinitrobenzene

CPO109

Quantitative Analysis

3. Amongst the following Kjeldahl's method can't be used for.

I. $PhNO_2$ II. $Ph-N=N-Ph$ III. $PhNHCOPh$ IV.  V. 

VI.  VII. $PhNH_2$ VIII. Acetone IX. Ethane nitrile X. Urea

CPO110

4. 0.50 g of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 mL of 0.5 M H_2SO_4. The residual acid required 60 mL of M/2 NaOH solution. Find the percentage of nitrogen in the compound.

CPO111

For More Join: @IITJEE_Advanced

5. 0.4 g of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 mL of 0.5 M H_3PO_3. The residual acid required 30 mL of 0.5 M $Ca(OH)_2$. Find the percentage of N_2 in the compound.

CPO112

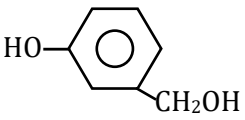
6. 0.002 g of an organic compound was treated according to Kjeldahl's method. 0.2×10^{-4} mol of H_2SO_4 was required to neutralize NH_3. Calculate the percentage of N

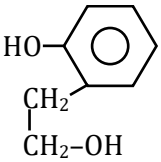
CPO113

Test of Functional Group

7. Amongst the following compounds the total number of compounds which gives coloured test with both ceric ammonium nitrate (CAN) and neutral $FeCl_3$ solution.

I. Benzyl alcohol II. Benzaldehyde III. Aniline

IV. 

V. 

VI. $PhCH_2CH(OH)CH_3$

VII. Urea VIII. Thiourea

CPO192

Part-2 : JEE Advanced (Decimal type)

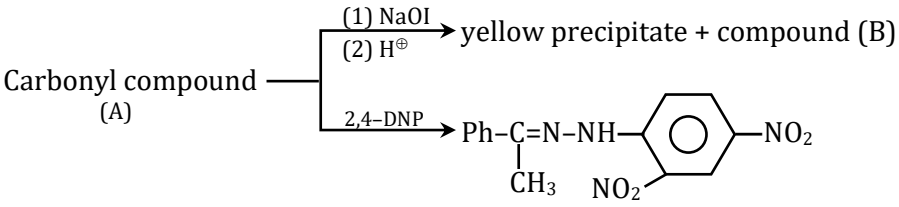
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Rankers Academy JEE

Qualitative Analysis

8. 0.12 g of an organic compound containing phosphorus gave 0.22 g of $Mg_2P_2O_7$ by the usual analysis. Calculate the percentage of phosphorus in the compound.

CPO114

9. 

Find the mass percentage of oxygen in compound (B).

CPO029

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Phenolic group can be identified by a positive: **[JEE (Main) 2024]**

(1) Phthalein dye test (2) Lucas test

(3) Tollen's test (4) Carbylamine test

CPO193

2. The technique used for purification of steam volatile water immiscible substance is: **[JEE (Main) 2024]**

(1) Fractional distillation

(2) Fractional distillation under reduced pressure

(3) Distillation

(4) Steam distillation

CPO194

3. Appearance of blood red colour, on treatment of the sodium fusion extract of an organic compound with $FeSO_4$ in presence of concentrated H_2SO_4 indicates the presence of element/s **[JEE (Main) 2024]**

(1) Br (2) N (3) N and S (4) S

CPO195

4. Chromatographic technique/s based on the principle of differential adsorption is/are

A. Column chromatography

B. Thin layer chromatography

C. Paper chromatography

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Choose the most appropriate answer from the options given below: **[JEE (Main) 2024]**

(1) B only (2) A only

(3) A & B only (4) C only

CPO196

5. Which among the following purification methods is based on the principle of "Solubility" in two different solvents? **[JEE (Main) 2024]**

(1) Column Chromatography (2) Sublimation

(3) Distillation (4) Differential Extraction

CPO197

6. Match List I with List II

List-I (Technique)		List-II (Application)	
A.	Distillation	I.	Separation of glycerol from spent-lye
B.	Fractional distillation	II.	Aniline - Water mixture
C.	Steam distillation	III.	Separation of crude oil fractions
D.	Distillation under reduced pressure	IV.	Chloroform-Aniline

Choose the correct answer from the options given below: **[JEE (Main) 2024]**

(1) A-IV, B-I, C-II, D-III (2) A-IV, B-III, C-II. D-I

(3) A-I. B-II, C-IV, D-III (4) A-II, B-III. C-I, D-IV

CPO198

For More Join: @IITJEE_Advanced

7. Given below are two statements :

Statement I: Aniline reacts with con. H_2SO_4 followed by heating at 453-473 K gives p-aminobenzene sulphonic acid, which gives blood red colour in the 'Lassaigne's test'.

Statement II: In Friedel - Craft's alkylation and acylation reactions, aniline forms salt with the $AlCl_3$ catalyst. Due to this, nitrogen of aniline acquires a positive charge and acts as deactivating group.

In the light of the above statements, choose the ***correct answer*** from the options given below :

[JEE (Main) 2024]

(1) Statement I is false but statement II is true
(2) Both statement I and statement II are false
(3) Statement I is true but statement II is false
(4) Both statement I and statement II are true

CPO199

8. The fragrance of flowers is due to the presence of some steam volatile organic compounds called essential oils. These are generally insoluble in water at room temperature but are miscible with water vapour in vapour phase. A suitable method for the extraction of these oils from the flowers is –

[JEE (Main) 2024]

(1) crystallisation
(2) distillation under reduced pressure
(3) distillation
(4) steam distillation

CPO200

9. Lassaigne's test is used for detection of : **[JEE (Main) 2024]**

(1) Nitrogen and Sulphur only
(2) Nitrogen, Sulphur and Phosphorous Only
(3) Phosphorous and halogens only
(4) Nitrogen, Sulphur, phosphorous and halogens

CPO201

Pankers Academy JEE

10. Which of the following nitrogen containing compound does not give Lassaigne's test ? **[JEE (Main) 2024]**

(1) Phenyl hydrazine (2) Glycene (3) Urea (4) Hydrazine

CPO202

11. The adsorbent used in adsorption chromatography is/are

A. silica gel B. alumina
C. quick lime D. magnesia

Choose the **most appropriate** answer from the options given below : **[JEE (Main) 2024]**

(1) B only (2) C and D only (3) A and B only (4) A only

CPO203

12. Ail organic compound has 42.1% carbon, 6.4% hydrogen and remainder is oxygen. If its molecular weight is 342, then its molecular formula is : **[JEE (Main) 2024]**

(1) $C_{11}H_{18}O_{12}$ (2) $C_{12}H_{20}O_{12}$ (3) $C_{14}H_{20}O_{10}$ (4) $C_{12}H_{22}O_{11}$

CPO204

13. The correct statements among the following, for a "chromatography" purification method is:

[JEE (Main) 2024]

(1) Organic compounds run faster than solvent in the thin layer chromatographic plate.
(2) Non-polar compounds are retained at top and polar compounds come down in column chromatography.
(3) R_f of a polar compound is smaller than that of a non-polar compound.
(4) R_f is an integral value.

CPO205

For More Join: @IITJEE_Advanced

14. Which of the following statements are correct ?

A. Glycerol is purified by vacuum distillation because it decomposes at its normal boiling point.

B. Aniline can be purified by steam distillation as aniline is miscible in water.

C. Ethanol can be separated from ethanol water mixture by azeotropic distillation because it forms azeotrope.

D. An organic compound is pure, if mixed M.P. is remained same.

Choose the **most appropriate** answer from the options given below : **[JEE (Main) 2024]**

(1) A, B, C only (2) A, C, D only (3) B, C, D only (4) A, B, D only

CPO206

15. Match **List-I** with **List-II**.

List-I (Test)		List-II (Identification)	
(A)	Bayer's test	(I)	Phenol
(B)	Ceric ammonium nitrate test	(II)	Aldehyde
(C)	Phthalein dye test	(III)	Alcoholic-OH group
(D)	Schiff's test	(IV)	Unsaturation

Choose the **correct** answer from the options given below : **[JEE (Main) 2024]**

(1) (A)-(III), (B)-(I), (C)-(IV), (D)-(II) (2) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(3) (A)-(IV), (B)-(I), (C)-(II), (D)-(III) (4) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

CPO207

16. Given below are two statements :

Statement (I) : Kjeldahl method is applicable to estimate nitrogen in pyridine.

Statement (II) : The nitrogen present in pyridine can easily be converted into ammonium sulphate in Kjeldahl method.

In the light of the above statements, choose the **correct** answer from the options given below.

[JEE (Main) 2024]

(1) Both **Statement I** and **Statement II** is false

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** is true

(4) **Statement I** is true but **Statement II** is false

CPO208

17. In Kjeldahl's method for estimation of nitrogen, $CuSO_4$ acts as : **[JEE (Main) 2024]**

(1) Reducing agent (2) Catalytic agent

(3) Hydrolysis agent (4) Oxidising agent

CPO209

18. Which of the following compounds will give silver mirror with ammoniacal silver nitrate?

(A) Formic acid (B) Formaldehyde

(C) Benzaldehyde (D) Acetone

Choose the correct answer from the options given below : **[JEE (Main) 2024]**

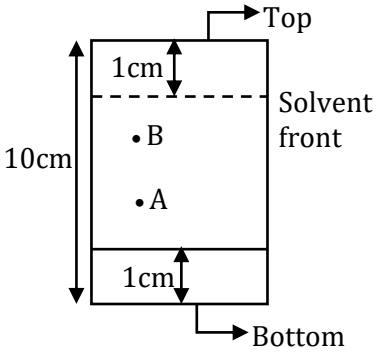
(1) C and D only (2) A, B and C only

(3) A only (4) B and C only

CPO210

For More Join: @IITJEE_Advanced

19.



In the given TLC, the distance of spot A & B are 5 cm & 7 cm, from the bottom of TLC plate, respectively. R_f value of B is $x \times 10^{-1}$ times more than A. The value of x is_____.

[JEE (Main) 2024]

CPO211

20. Using the given figure, the ratio of R_f values of sample A and sample C is $x \times 10^{-2}$. Value of x is _____. **[JEE (Main) 2024]**

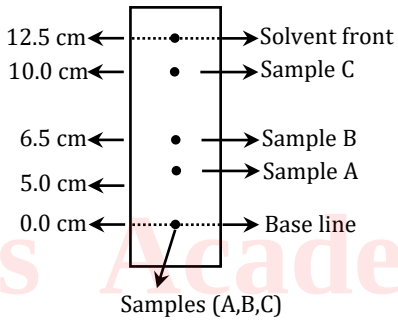


Fig : Paper chromatography of Samples

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CPO212

21. Methods used for purification of organic compounds are based on : **[JEE (Main) 2024]**

(1) neither on nature of compound nor on the impurity present.

(2) nature of compound only.

(3) nature of compound and presence of impurity.

(4) presence of impurity only.

CPO213

22. In sulphur estimation. 0.471 g of an organic compound gave 1.4439 g of barium sulphate.

The percentage of sulphur in the compound is_____(Nearest Integer)

(Given: Atomic mass Ba: 137 u: S: 32 u, O: 16 u) **[JEE (Main) 2023]**

CPO119

23. Compound that will give positive Lassaigne's test for both nitrogen and halogen is

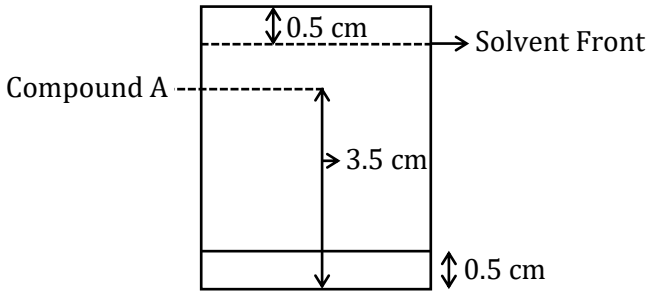
[JEE (Main) 2023]

(1) $N_2H_4.HCl$ (2) $CH_3NH_2.HCl$ (3) NH_4Cl (4) $NH_2OH.HCl$

CPO120

For More Join: @IITJEE_Advanced

24. Following chromatogram was developed by adsorption of compound 'A' on a 6 cm TLC glass plate. Retardation factor of the compound 'A' is _____ × 10^{-1}. **[JEE (Main) 2023]**



CPO121

25. When 0.01 mol of an organic compound containing 60% carbon was burnt completely, 4.4 g of CO_2 was produced. The molar mass of compound is _____ g mol^{-1} (Nearest integer)

[JEE (Main) 2023]

CPO122

26. Match List I with List II: **[JEE (Main) 2023]**

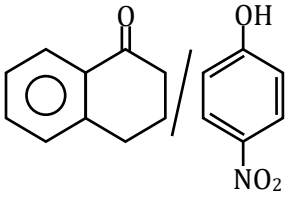
List I (Mixture)		**List II (Separation Technique)**	
(A)	$CHCl_3 + C_6H_5NH_2$	I	Steam distillation
(B)	$C_6H_{14} + C_5H_{12}$	II	Differential extraction
(C)	$C_6H_5NH_2 + H_2O$	III	Distillation
(D)	Organic compound in H_2O	IV	Fractional distillation

(1) A-IV, B-I, C-III, D-II (2) A-III, B-IV, C-I, D-II
(3) A-II, B-I, C-III, D-IV (4) A-III, B-I, C-IV, D-II

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CPO123

27. Match items of column I and II **[JEE (Main) 2023]**

	Column I (Mixture of compounds)		**Column II (Separation Technique)**
A.	H_2O/CH_2Cl_2	i.	Crystallization
B.		ii.	Differential solvent extraction
C.	Kerosene/Naphthalene	iii.	Column chromatography
D.	$C_6H_{12}O_6$/NaCl	iv.	Fractional Distillation

Correct match is :
(1) A-(iii), B-(iv), C-(ii), D-(i) (2) A-(i), B-(iii), C-(ii), D-(iv)
(3) A-(ii), B-(iii), C-(iv), D-(i) (4) A-(ii), B-(iv), C-(i), D-(iii)

CPO124

28. On complete combustion, 0.492 g of an organic compound gave 0.792 g of CO_2. The % of carbon in the organic compound is ____ (Nearest integer) **[JEE (Main) 2023]**

CPO125

For More Join: @IITJEE_Advanced

29. In Dumas method for the estimation of N_2, the sample is heated with copper oxide and the gas evolved is passed over : **[JEE (Main) 2023]**

(1) Ni (2) Copper gauze (3) Pd (4) Copper oxide

CPO126

30. **Match List I with List II** **[JEE (Main) 2023]**

List I Element detected		List II Reagent used/Product formed	
A	Nitrogen	I.	$Na_2[Fe(CN)_5 NO]$
B	Sulphur	II.	$AgNO_3$
C	Phosphorous	III.	$Fe_4[Fe(CN)_6]_3$
D	Halogen	IV.	$(NH_4)_2 MoO_4$

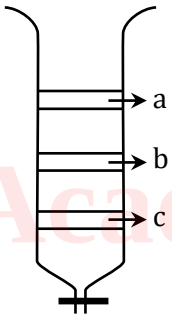
Choose the correct answer from the options given below:

(1) A-II, B-IV, C-I, D-III (2) A-IV, B-II, C-I, D-III

(3) A-II, B-I, C-IV, D-III (4) A-III, B-I, C-IV, D-II

CPO127

31. From the figure of column chromatography given below, identify incorrect statements. **[JEE (Main) 2023]**



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A. Compound 'c' is more polar than 'a' and 'b'

B. Compound 'a' is least polar

C. Compound 'b' comes out of the column before 'c' and after 'a'

D. Compound 'a' spends more time in the column

Choose the correct answer from the options given below :-

(1) A, B and C only (2) B, C and D only

(3) A, B and D only (4) B and D only

CPO128

32. Using column chromatography, mixture of two compounds 'A' and 'B' was separated. 'A' eluted first, this indicates 'B' has **[JEE (Main) 2023]**

(1) low R_f, weaker adsorption (2) high R_f, stronger adsorption

(3) high R_f, weaker adsorption (4) low R_f, stronger adsorption

CPO129

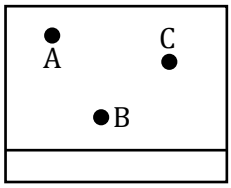
33. In Carius tube, an organic compound 'X' is treated with sodium peroxide to form a mineral acid 'Y'. The solution of $BaCl_2$ is added to 'Y' to form a precipitate 'Z'. 'Z' is used for the quantitative estimation of an extra element. 'X' could be: **[JEE (Main) 2023]**

(1) Cytosine (2) Chloroxylenol (3) A nucleotide (4) Methionine

CPO130

For More Join: @IITJEE_Advanced

34. Thin layer chromatography of a mixture shows the following observation :



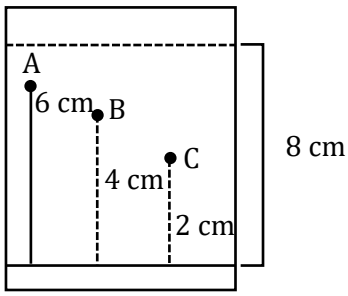
The correct order of elution in the silica gel column chromatography is **[JEE (Main) 2023]**

(1) A, C, B (2) B, C, A (3) C, A, B (4) B, A, C

CPO131

35. Three organic compounds A, B and C were allowed to run in thin layer chromatography using hexane and gave the following result (see figure). The R_f value of the most polar compound is ___$\times 10^{-2}$

[JEE (Main) 2023]



CPO132

36. An organic compound gives 0.220 g of CO_2 and 0.126 g of H_2O on complete combustion. If the % of carbon is 24 then the % hydrogen is _____ $\times 10^{-1}$. (Nearest integer) **[JEE (Main) 2023]**

CPO133

37. 0.400 g of an organic compound (X) gave 0.376 g of AgBr in Carius method for estimation of bromine. % of bromine in the compound (X) is____.

Given: Molar mass AgBr = 188 g mol^{-1} Br = 80 g mol^{-1}) **[JEE (Main) 2023]**

CPO134

38. Which of the following statement is correct for paper chromatography ? **[JEE (Main) 2023]**

(1) Water present in the mobile phase gets absorbed by the paper which then forms the stationary phase.

(2) Water present in the pores of the paper forms the stationary phase.

(3) Paper sheet forms the stationary phase.

(4) Paper and water present in its pores together form the stationary phase.

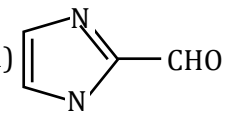
CPO135

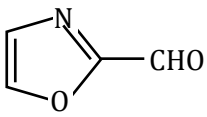
39. Which of the following compounds would give the following set of qualitative analysis ?

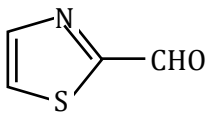
(i) Fehling's Test : Positive

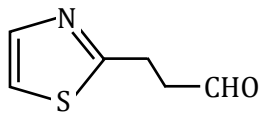
(ii) Na fusion extract upon treatment with sodium nitroprusside gives a blood red colour but not

[JEE (Main) 2023]

(1) 

(2) 

(3) 

(4) 

CPO136

For More Join: @IITJEE_Advanced

40. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R.**
Assertion A : Thin layer chromatography is an adsorption chromatography.
Reason R : A thin layer of silica gel is spread over a glass plate of suitable size in thin layer chromatography which acts as an adsorbent.
In the light of the above statements, choose the ***correct*** answer from the options given below
[JEE (Main) 2022]

(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is NOT the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

CPO117

41. A sample of 0.125 g of an organic compound when analysed by Duma's method yields 22.78 mL of nitrogen gas collected over KOH solution at 280K and 759 mm Hg. The percentage of nitrogen in the given organic compound is ___. (Nearest integer). **[JEE (Main) 2022]**
Given:
(A) The vapour pressure of water at 280 K is 14.2 mm Hg
(B) R = 0.082 L atm K^{-1} mol^{-1}

CPO118

42. Match List-I with List-II **[JEE (Main) 2021]**

	List-I Test/Reagents/Observation(s)		**List-II Species detected**
(a)	Lassaigne's Test	(i)	Carbon
(b)	Cu(II) oxide	(ii)	Sulphur
(c)	Silver nitrate	(iii)	N, S, P, and halogen
(d)	The sodium fusion extract gives black precipitate with acetic acid and lead acetate	(iv)	Halogen Specifically

The correct match is :
(1) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
(2) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
(3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
(4) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

CPO116

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EXERCISE - JEE (Advanced) PYQ

1. Match the compounds in List-I with the observation in List-II, and choose the correct option.

	List-I		List-II
(I)	Aniline	(P)	Sodium fusion extract of the compound on boiling with $FeSO_4$, followed by acidification with conc. H_2SO_4, gives Prussian blue color.
(II)	o-Cresol	(Q)	Sodium fusion extract of the compound on treatment with sodium nitroprusside gives blood red color.
(III)	Cysteine	(R)	Addition of the compound to a saturated solution of $NaHCO_3$ results in effervescence.
(IV)	Coprolactam	(S)	The compound reacts with bromine water to give a white precipitate.
		(T)	Treating the compound with neutral $FeCl_3$ solution produces violet colour.

[JEE (Advanced) 2022]

(A) I → P, Q; II → S; III → Q, R; IV → P
(B) I → P ; II → R, S; III → R; IV → Q, S
(C) I → Q, S; II → P, T; III → P; IV → S
(D) I → P, S; II → T; III → Q, R; IV → P

CPO067

2. For the identification of β-naphthol using dye test, it is necessary to use **[JEE (Advanced) 2014]**

(A) dichloromethane solution of β-naphthol
(B) acidic solution of β-naphthol
(C) neutral solution of β-naphthol
(D) alkaline solution of β-naphthol

CPO060

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	D	C	B	A	B	B	D	B
Que.	11	12	13	14	15	16	17	18	19	
Ans.	C	D	D	C	C	A	C	B	C	

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABCD	BCD	ABC	AB	BC	BC	AB	C	B	D
Que.	11	12	13	14						
Ans.	A	C	A	B						

EXERCISE (S)

1. 3 **2.** 2 **3.** 6 **4.** 56 **5.** 70
6. 28 **7.** 2 **8.** 51.2 **9.** 26.22

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	4	3	3	4	2	4	4	4	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	4	3	2	4	1	2	2	15	50
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	42	2	6	200	2	3	44	2	4
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	1	4	4	1	25	56	40	2	4	1
Que.	41	42								
Ans.	22	3								

EXERCISE - JEE (Advanced) PYQ

Que.	1	2
Ans.	D	D

For More Join: @IITJEE_Advanced

Important Notes

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For More Join: @IITJEE_Advanced